Bio-Neighbor Treatment Auditor Report

Generated 2026-05-07T18:33:01Z
Source: https://github.com/greenrobotllc/bio-neighbor

Research tool only — not medical advice. This report compiles publicly available data from NCI PDQ and ClinicalTrials.gov plus on-device LLM summaries. Discuss any treatment-plan questions with your oncology team.

1. Inputs

Cancer type	Breast
Subtype	HER2+
Markers	HER2+, ERBB2
Stage	Stage IV
Stage detail	metastasized to bone in the hip
Prescribed drugs	Anastrozole (CHEMBL1399)
Scheduled treatments	(none)
Symptoms / side effects	(none)

2. Methodology & data sources

Multi-pass synthesis

The audit is intentionally split into many small LLM calls instead of one giant prompt — each source is summarized in isolation so a large source can't drown the others out, and the final synthesis reasons over the digests rather than the raw data:

1. **Source fetch.** PDQ summary (1 call), modality trials (4 parallel CT.gov queries), per-drug trials (1 parallel CT.gov queries).
2. **Per-source mini-summaries.** Each non-empty source is run through Ollama with the patient plan as context, producing a 120–180 word digest pointed at this patient's subtype/stage. 6 mini-summaries produced for this run.
3. **Final synthesis.** A single Ollama call combines the patient plan + every mini-summary into

the 350–550 word audit shown below, with explicit NCT-ID citations and a "Further reading" block. The raw source data is not in the synthesis prompt — the model only sees the mini-summaries it produced earlier.

This pipeline is fully reproducible by hand: pull the same PDQ page, run the same CT.gov queries listed above, then paste each source into an LLM with the patient plan and ask for the same digests.

NCI PDQ — Health Professional

URL fetched: https://www.cancer.gov/types/breast/hp/breast-treatment-pdq

Filtered for stage: `Stage IV` Stage-detail tokens scored against headings + body: `metastasized to bone in the hip`.

The page is parsed for `h2/h3/h4 + p/li`; sections are scored against (stage keywords, marker keywords, stage-detail tokens) with always-include for "stage information", "treatment option overview", "surgical treatment", "radiation therapy". Top-N kept, capped at 6 sections / 6,000 chars total. Sections retained for this run:

- `Metastatic HER2-Low Breast Cancer` (h3)
- `Metastatic HER2-Positive Breast Cancer` (h3)
- `Bone-Modifying Therapy for Metastatic Breast Cancer` (h3)
- `Treatment Option Overview for Metastatic Breast Cancer` (h3)
- `Surgical Treatment for Metastatic Breast Cancer` (h3)
- `Radiation Therapy for Metastatic Breast Cancer` (h3)

ClinicalTrials.gov — modality search

For each of the four modalities (radiation, surgery, chemotherapy, targeted), a parallel CT.gov v2 query is run via `GET /studies` with these parameters:

- `query.cond` = subtype's first ChEMBL indication term if present, else the subtype name (this run: `HER2+`; the backend resolves the exact term server-side).
- `query.intr` = modality intervention term (see table below).
- `filter.overallStatus = COMPLETED|TERMINATED|ACTIVE_NOT_RECRUITING` — completed trials with reportable outcomes are preferred over actively-recruiting trials.
- `pageSize` = 100, up to 3 pages, hard-capped at 300 raw records before sort.
- Sort: trials with multi-arm reported numeric outcomes first (apples-to-apples comparisons), then any-results, then more arms, with NCT-ID as a stable tiebreak.
- No date-range filter is applied; ordering does the prioritization.
- Up to 8 trials per modality are returned to the audit pass.

Modality	CT.gov intervention term	Result
`radiation`	`query.intr=radiation therapy`	8 trials returned

`surgery`	`query.intr=surgery`	8 trials returned
`chemotherapy`	`query.intr=chemotherapy`	8 trials returned
`targeted`	`query.intr=targeted therapy`	8 trials returned

ClinicalTrials.gov — per-drug trials

For each prescribed drug with a ChEMBL ID, NCT IDs are pulled from ChEMBL's `drug_indication` records, then each trial is fetched from CT.gov v2 (`GET /studies/<NCT>`) in parallel (max 5 concurrent). The same multi-arm-results-first sort is applied. Up to 10 trials per drug are surfaced.

Drug	ChEMBL ID	Result
Anastrozole	`CHEMBL1399`	10 trials returned

3. Audit pipeline log

Live trace of fetches and LLM passes for this run.

State	Step	Detail
done	Pulling NCI PDQ for Breast	
done	Searching radiation trials	
done	Searching surgery trials	
done	Searching chemotherapy trials	
done	Searching targeted trials	
done	Fetching trials for 1 drug	
done	Summarizing NCI PDQ — Breast	
done	Summarizing Radiation trials	
done	Summarizing Surgery trials	
done	Summarizing Chemotherapy trials	
done	Summarizing Targeted trials	
done	Summarizing Anastrozole trials	
done	Synthesizing final audit	

4. Per-source summaries

NCI PDQ — Breast

The NCI PDQ source is highly relevant to this patient’s Stage IV HER2+ breast cancer with bone metastasis. Under the "Treatment of Metastatic Breast Cancer" section, the source identifies several approved HER2-targeted therapies, such as trastuzumab, pertuzumab, trastuzumab emtansine, and lapatinib, which are not included in the patient's current Anastrozole regimen. To address the patient's hip metastasis, the "Bone-Modifying Therapy" section recommends using agents like pamidronate, clodronate, or zoledronate to reduce skeletal morbidity, specifically citing trial NCT00869206 regarding zoledronic acid dosing. While the patient's use of Anastrozole aligns with the "Treatment Option Overview" regarding aromatase inhibitors, the plan notably lacks the specified HER2-targeted agents and bone-modifying therapies. Furthermore, the source discusses radiation therapy for symptomatic palliation and locoregional interventions (NCT01242800), neither of which are currently part of the patient's prescribed plan.

Radiation trials

This source contains several trials relevant to the patient's HER2-positive, metastatic (Stage IV) profile. For metastatic disease, NCT03939897 explores the addition of Copanlisib to Abemaciclib and Fulvestrant. In trials focused on brain metastases, NCT00437073 reported five participants with CNS responses using Lapatinib plus Capecitabine, while NCT01622868 reported a 5.8% complete response rate for radiation therapy alone compared to 0% when combined with Lapatinib. The patient’s current plan using Anastrozole does not include the various targeted therapies or radiation modalities discussed in these trials, such as CDK4/6 inhibitors (Abemaciclib), PI3K inhibitors (Copanlisib), or HER2-targeted agents (Trastuzumab, Pertuzumab, and Lapatinib). Additionally, the source details radiation-based interventions like Whole-Brain Radiotherapy (WBRT) and Stereotactic Radiosurgery (SRS) that are absent from the current treatment plan.

Surgery trials

The provided trials include data pertinent to HER2+ breast cancer. While much of the research focuses on early or locally advanced stages, NCT01494662 addresses HER2+ metastatic disease, noting a 48.6% objective response rate for Capecitabine (Cohort 3A). Regarding administration, NCT05415215 found that 69% of participants in Arm C preferred the home administration of subcutaneous Pertuzumab and Trastuzumab following hospital-based treatment. Additional trials, such as NCT01008150 and NCT02789657, investigate neoadjuvant HER2+ regimens using agents like Neratinib, Paclitaxel, and Carboplatin. Notably, the patient's current plan of Anastrozole does not include the HER2-targeted therapies or chemotherapy classes identified in these studies, such as Trastuzumab, Pertuzumab, Neratinib, Trastuzumab Emtansine, Capecitabine, or Paclitaxel.

Chemotherapy trials

The source includes trials relevant to HER2+ and metastatic breast cancer, particularly regarding resistance to aromatase inhibitors, the class of the patient's Anastrozole. In NCT02028507, which studied metastatic breast cancer resistant to aromatase inhibitors, Capecitabine demonstrated a progression-free survival of 9.4 months, compared to 7.3 months for Palbociclib plus Exemestane. Regarding HER2+ therapies, NCT00490139 found a 1.9-year disease-free survival with Lapatinib plus Trastuzumab, and NCT00524303 reported a 74% pathological complete response rate for a Trastuzumab and Lapatinib combination. The patient's plan lacks several drug classes and modalities mentioned in the source, including other HER2-targeted agents (Pertuzumab, Trastuzumab Emtansine, and Lapatinib), chemotherapy combinations (including Docetaxel, Carboplatin, and Paclitaxel), and CDK4/6 inhibitors like Palbociclib.

Targeted trials

This source contains several trials relevant to the patient’s HER2+ subtype. Specifically, NCT00021255 reports 5-year disease-free survival rates between 75.5% and 83.2% for various regimens involving Herceptin and chemotherapy. Regarding safety, NCT00976989 evaluates Pertuzumab and Trastuzumab combined with chemotherapy, noting low rates of symptomatic cardiac events (0% to 2.7%) and LVEF decline. Additionally, NCT03321981 examines Zenocutuzumab in combination with Trastuzumab and chemotherapy or endocrine therapy. While the patient’s current plan uses Anastrozole, the HER2+ trials in this source focus predominantly on HER2-targeted monoclonal antibodies—such as Trastuzumab, Pertuzumab, and Zenocutuzumab—and chemotherapy combinations including Docetaxel, Carboplatin, and Paclitaxel. The source does not address the use of aromatase inhibitors for HER2+ disease, prioritizing instead targeted agents and cytotoxic-based regimens.

> **Anastrozole trials**
>
> The source provides clinical trial data relevant to HER2-positive metastatic breast cancer, specifically within NCT03179904. This trial investigates combinations involving the FASN inhibitor TVB-2640, trastuzumab, and endocrine therapies including anastrozole. A specific finding worth noting is that Cohort A, which used denifanstat, paclitaxel, and trastuzumab, demonstrated an Overall Response Rate of 0.333, while the rate for Cohort B, involving an endocrine therapy regimen with anastrozole, was not available (NA). Additionally, NCT04256941 examines aromatase inhibitor therapy for HR-positive metastatic breast cancer with ERS1 mutations. Notably, the source mentions several therapeutic modalities and drug classes that are absent from the patient's current plan, including targeted therapies such as trastuzumab and denifanstat, chemotherapy like paclitaxel, and other endocrine or CDK4/6 inhibitors including fulvestrant, letrozole, exemestance, abemaciclib, palbociclib, and ribociclib.

5. Final audit

Treatment Plan Audit

1. Efficacy signals
The current plan relies on Anastrozole, an aromatase inhibitor. While the provided summaries do not present direct evidence of Anastrozole's efficacy specifically for the HER2+ subtype, they highlight strong positive signals for HER2-targeted combinations. For example, NCT00490139 reported a 1.9-year disease-free survival using Lapatinib and Trastuzumab, and NCT00524303 showed a 74% pathological complete response rate with a similar combination.

2. Alternative or adjunct regimens
Some trials suggest that certain modalities may alter outcomes compared to drug-only approaches. In NCT01622868, data indicated a 5.8% complete response rate for radiation therapy alone, compared to 0% when combined with Lapatinib. Regarding administration, NCT054155215 noted that 69% of participants in a specific study arm preferred the home-based subcutaneous administration of Pertuzumab and Trastuzumab over hospital-based treatment.

3. Symptom & side-effect concerns
The patient is experiencing bone metastasis in the hip. While the current plan does not include bone-modifying agents, the NCI PDQ suggests these are standard for reducing skeletal issues. Additionally, if the treatment plan is updated to include HER2-targeted therapies like Trastuzumab or Pertuzumab, the patient should be monitored for cardiac side effects, as NCT00976989 noted risks of symptomatic cardiac events and LVEF (left ventricular ejection fraction) decline.

4. Plan gaps
Based on the NCI PDQ (https://www.cancer.gov/types/breast/hp/breast-treatment-pdq), the current plan lacks several standard-of-care components for HER2+ metastatic disease.

Specifically, the plan is missing HER2-targeted therapies such as trastuzumab, pertuzumab, trastuzumab emtansine, or lapatinib. Furthermore, to address the hip metastasis, the plan lacks bone-modifying therapies like zoledronate (NCT00869206), clodronate, or pamidronate. The plan also does not include radiation therapy for symptom relief (NCT01242800). You should discuss these potential additions with your oncology team.

5. Uncertainty
There is thin evidence in the provided summaries regarding the effectiveness of Anastrozole specifically for the HER2+ subtype, as most high-efficacy trials focus on monoclonal antibodies and chemotherapy. I would ask the oncology team: "Given the HER2+ status of this cancer, why is an aromatase inhibitor being used instead of HER2-targeted agents like Trastuzumab or Pertuzumab?"

***

Further reading
- NCI PDQ summary: https://www.cancer.gov/types/breast/hp/breast-treatment-pdq
- https://clinicaltrials.gov/study/NCT00490139
- https://clinicaltrials.gov/study/NCT01622868
- https://clinicaltrials.gov/study/NCT00869206

Not medical advice. Discuss any plan changes with your oncologist.

6. References

NCI PDQ — Breast (Health Professional): https://www.cancer.gov/types/breast/hp/breast-treatment-pdq

ClinicalTrials.gov studies cited / surfaced (35):

- `NCT05415215` (Radiation trial): https://clinicaltrials.gov/study/NCT05415215 — A Study to Evaluate Patient Preference for Home Administration of Fixed-Dose Combination of Pertuzumab and Trastuzumab for Subcutaneous Administration in Participants With Early or Locally Advanced/Inflammatory HER2-Positive Breast Cancer
- `NCT03939897` (Radiation trial): https://clinicaltrials.gov/study/NCT03939897 — Testing the Addition of Copanlisib to Usual Treatment (Fulvestrant and Abemaciclib) in Metastatic Breast Cancer
- `NCT00437073` (Radiation trial): https://clinicaltrials.gov/study/NCT00437073 — Brain Metastases In ErbB2-Positive Breast Cancer
- `NCT00769379` (Radiation trial): https://clinicaltrials.gov/study/NCT00769379 — Radiation Therapy With or Without Trastuzumab in Treating Women With Ductal Carcinoma In Situ Who Have Undergone Lumpectomy
- `NCT01217411` (Radiation trial): https://clinicaltrials.gov/study/NCT01217411 — RO4929097 and Whole-Brain Radiation Therapy or Stereotactic Radiosurgery in Treating Patients With Brain Metastases From Breast Cancer

- `NCT01622868` (Radiation trial): https://clinicaltrials.gov/study/NCT01622868 — Whole-Brain Radiation Therapy or Stereotactic Radiosurgery With or Without Lapatinib Ditosylate in Treating Patients With Brain Metastasis From HER2-Positive Breast Cancer
- `NCT02003209` (Radiation trial): https://clinicaltrials.gov/study/NCT02003209 — Docetaxel, Carboplatin, Trastuzumab, and Pertuzumab With or Without Estrogen Deprivation in Treating Patients With Hormone Receptor-Positive, HER2-Positive Operable or Locally Advanced Breast Cancer
- `NCT02132949` (Radiation trial): https://clinicaltrials.gov/study/NCT02132949 — A Study Evaluating Pertuzumab (Perjeta) Combined With Trastuzumab (Herceptin) and Standard Anthracycline-based Chemotherapy in Participants With Human Epidermal Growth Factor Receptor 2 (HER2)-Positive Locally Advanced, Inflammatory, or Early-stage Breast Cancer
- `NCT04588298` (Surgery trial): https://clinicaltrials.gov/study/NCT04588298 — A Study to Investigate the Biological Effects of AZD9833 in Women With ER-positive, HER2 Negative Primary Breast Cancer
- `NCT01008150` (Surgery trial): https://clinicaltrials.gov/study/NCT01008150 — Phase II Randomized Trial Evaluating Neoadjuvant Therapy With Neratinib and/or Trastuzumab Followed by Postoperative Trastuzumab in Women With Locally Advanced HER2-positive Breast Cancer
- `NCT01494662` (Surgery trial): https://clinicaltrials.gov/study/NCT01494662 — HKI-272 for HER2-Positive Breast Cancer and Brain Metastases
- `NCT02520063` (Surgery trial): https://clinicaltrials.gov/study/NCT02520063 — Combination of Letrozole, Everolimus and TRC105 in Postmenopausal Women With Hormone-Receptor Positive and Her2 Negative Breast Cancer
- `NCT02789657` (Surgery trial): https://clinicaltrials.gov/study/NCT02789657 — Neoadjuvant Therapy in Clinical Stage I-III HER2-positive Breast Cancer.
- `NCT00045032` (Surgery trial): https://clinicaltrials.gov/study/NCT00045032 — Herceptin (Trastuzumab) in Treating Women With Human Epidermal Growth Factor Receptor (HER) 2-Positive Primary Breast Cancer
- `NCT00490139` (Chemotherapy trial): https://clinicaltrials.gov/study/NCT00490139 — ALTTO (Adjuvant Lapatinib And/Or Trastuzumab Treatment Optimisation) Study; BIG 2-06/N063D
- `NCT02028507` (Chemotherapy trial): https://clinicaltrials.gov/study/NCT02028507 — Phase III Palbociclib With Endocrine Therapy vs. Capecitabine in HR+/HER2- MBC With Resistance to Aromatase Inhibitors
- `NCT00021255` (Chemotherapy trial): https://clinicaltrials.gov/study/NCT00021255 — Combination Chemotherapy With or Without Trastuzumab in Treating Women With Breast Cancer
- `NCT00524303` (Chemotherapy trial): https://clinicaltrials.gov/study/NCT00524303 — Lapatinib +/- Trastuzumab In Addition To Standard Neoadjuvant Breast Cancer Therapy.
- `NCT00976989` (Chemotherapy trial): https://clinicaltrials.gov/study/NCT00976989 — A Study of Pertuzumab in Combination With Herceptin and Chemotherapy in Participants With HER2-

Positive Breast Cancer

- NCT01009918 (Chemotherapy trial): https://clinicaltrials.gov/study/NCT01009918 — Lisinopril or Coreg CR® in Reducing Side Effects in Women With Breast Cancer Receiving Trastuzumab
- NCT05101564 (Targeted trial): https://clinicaltrials.gov/study/NCT05101564 — Umbrella Trial of Subtype-Targeted Therapies in ER+/HER2- Breast Cancer
- NCT03056755 (Targeted trial): https://clinicaltrials.gov/study/NCT03056755 — Study Assessing the Efficacy and Safety of Alpelisib Plus Fulvestrant or Letrozole, Based on Prior Endocrine Therapy, in Patients With PIK3CA Mutant, HR+, HER2- Advanced Breast Cancer Who Have Progressed on or After Prior Treatments
- NCT03321981 (Targeted trial): https://clinicaltrials.gov/study/NCT03321981 — MCLA-128 With Trastuzumab/Chemotherapy in HER2+ and With Endocrine Therapy in ER+ and Low HER2 Breast Cancer.
- NCT03747120 (Targeted trial): https://clinicaltrials.gov/study/NCT03747120 — Neoadjuvant Her2-targeted Therapy and Immunotherapy With Pembrolizumab [IIT2018-04-MCARTHUR-NEOHP]
- NCT00968968 (Targeted trial): https://clinicaltrials.gov/study/NCT00968968 — Continued HER2 Suppression With Lapatinib Plus Trastuzumab Versus Trastuzumab Alone
- NCT02441946 (Anastrozole trial): https://clinicaltrials.gov/study/NCT02441946 — A Neoadjuvant Study of Abemaciclib (LY2835219) in Postmenopausal Women With Hormone Receptor Positive, HER2 Negative Breast Cancer
- NCT01545336 (Anastrozole trial): https://clinicaltrials.gov/study/NCT01545336 — Anastrozole in Patients With Pulmonary Arterial Hypertension
- NCT03179904 (Anastrozole trial): https://clinicaltrials.gov/study/NCT03179904 — TVB-2640 and Trastuzumab With Paclitaxel or Endocrine Therapy for Treatment of HER2 Positive Metastatic Breast Cancer
- NCT03229499 (Anastrozole trial): https://clinicaltrials.gov/study/NCT03229499 — Pulmonary Hypertension and Anastrozole Trial
- NCT04256941 (Anastrozole trial): https://clinicaltrials.gov/study/NCT04256941 — Aromatase Inhibitor Therapy With or Without Fulvestrant for the Treatment of HR Positive Metastatic Breast Cancer With an ERS1 Activating Mutation, the INTERACT Study
- NCT00661531 (Anastrozole trial): https://clinicaltrials.gov/study/NCT00661531 — Estrogen in Postmenopausal Women With ER Positive Metastatic Breast Cancer After Failure of Sequential Endocrine Therapy
- NCT01394211 (Anastrozole trial): https://clinicaltrials.gov/study/NCT01394211 — Neo-adjuvant Therapy With Anastrozole Plus Pazopanib in Stage II and III ER+ Breast Cancer
- NCT01776008 (Anastrozole trial): https://clinicaltrials.gov/study/NCT01776008 — Akt Inhibitor MK-2206 and Anastrozole With or Without Goserelin Acetate in Treating Patients With Stage II-III Breast Cancer
- NCT04294225 (Anastrozole trial): https://clinicaltrials.gov/study/NCT04294225 — Anastrozole and Letrozole After Surgery for the Treatment of Stage I-III Breast Cancer

- `NCT05501704` (Anastrozole trial): https://clinicaltrials.gov/study/NCT05501704 — ETHAN - ET for Male BC

Bio-Neighbor source & methodology: https://github.com/greenrobotllc/bio-neighbor

---

Generated by Bio-Neighbor Treatment Auditor on 2026-05-07T18:33:01Z.

Repository: https://github.com/greenrobotllc/bio-neighbor